

Jack Rollo

DA 401

Dr. Wang

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The MLB Home Run Derby is a highly followed event that takes place in mid-July, during the All-Star Break of the current MLB season. For decades, fans and critics have wondered if a player participating in the Derby contributes to a decline in his performance throughout the remainder of the season. Participating in the Derby entails taking several high-intent swings in which a player is solely trying to hit a home run (DeFroda et. al, 2021), which is much different than his usual approach when preparing for a game. This has led to a handful of research in recent years to determine if there is a statistically significant decline in performance among participants in the 2nd half of the season. Most studies have found the decline to be insignificant or a myth altogether. However, what these studies have failed to do is hone in on specific players and their batted-ball results before and after the Derby. Having success in the Derby typically relies on pulling the ball in the air, which requires hitting the ball sooner on its flight to the plate, potentially changing the hitter's swing mechanics in a fashion that will affect his performance the rest of the season. As I carry experience in baseball as both a player and analyst, I am confident in my ability to use past data to gather insights that have not been questioned before. A research paper will be the best output for my work, as it allows my ideas and insights to be shown in the clearest manner. Some tools I plan to incorporate include multivariate regression, confidence intervals, and t-tests, in order to evaluate changes between 1st and 2nd-half performance (Forman, 2019). I already have a strong understanding of k-means clustering, which will be useful to define different types of hitter approaches, and I plan to expand my knowledge of PCA, which will be extremely helpful in evaluating the consistency and importance of offensive statistics that I will use for comparison throughout the project.

In the initial stages of my research, it will be important to refine my understanding of regression modeling, confidence intervals, and t-tests. While these are foundational and somewhat simple methods in

data analysis, they are nonetheless important to have a strong handle on. I learned these throughout my early data analytics classes, mainly in DA220, before using them much more frequently throughout my time working at Premier Pitching Performance (P3) in the summers of 2025 and 2026. It was at P3 where I first gathered the concept of k-means clustering, which was also incorporated into my DA352 class. A skill I am familiar with but currently unable to explain with confidence is Principal Component Analysis. I have seen my peers use it in projects of their own, but I have not used it in a professional setting. However, I know it is effective in cleaning and simplifying large portions of data (Frost, 2023), which I know is a task I will face in the early stages of my research.

I have watched over 10 different Home Run Derbies, and followed over 15 MLB seasons in my time as a baseball fan. Completing a project such as this one requires the knowledge and context received from consistently following both the event and the sport as a whole, with an ability to evaluate players and understand the multitude of factors that can influence performance. On the analysis side, I also need a strong foundation in statistical reasoning, correlation vs causation, selection bias, and an understanding of statistical evidence and what may influence it. Given the background I have as a baseball fan and analyst, I feel I am well prepared to take on these tasks. The current scholarly literature provides plenty of prior research that proposes questions about inflicted injuries and 2nd half declines. However, there is very little research that answers questions about changes in approach and batted ball data, in addition to scarce work published in the past 5 years. With recent changes in the Home Run Derby format (Major League Baseball, 2026), this presents an opportunity for me to create my own twist and challenge additional aspects of the Home Run Derby's effect that have not been publicly answered before.

Initial Proposal	9/13
All References Collected	9/15
Literature Review	9/20

All Data from All Sources Collected/Cleaned	9/23
Research Design Draft 1	9/27
Revised Research Design	10/4
Arrange conversations with non-sports (or non-analytics) audience to ensure scope and research is clear so far	10/12
Present Early Results	10/19
First Draft of Final Paper	11/2
Present to Family Members	11/22-11/29 (Thanksgiving Break)
Executive Summary	11/30
Paper Final Draft	12/7
Poster Presentation	12/12

I believe using the table above as a guideline for myself throughout this project will elicit the best results. In addition to the required assignments in the class curriculum throughout the semester, I have included multiple instances where I present to non-DA or non-sports audiences to get a better understanding of how I convey my ideas.

Works Cited

DeFroda, S. F., Staffa, S. J., Keeley, T., & Kriz, P. K. (2021). Home Run Derby participation in Major League Baseball players: Is there associated injury risk and impact on second-half performance? *Orthopaedic Journal of Sports Medicine*, 9(2), 1–8. <https://doi.org/10.1177/2325967120983350>

Forman, S. (2019, January 9). *We were wrong about the Home Run Derby curse*. FanGraphs Community Blog.

Frost, J. (2023). *Principal component analysis guide & example*. Statistics By Jim.

<https://statisticsbyjim.com/basics/principal-component-analysis/>

Major League Baseball. (2026, June 18). *Changes coming to 2026 Home Run Derby*. MLB.com.

<https://www.mlb.com/news/home-run-derby-format-change-for-2026>